

COSC264 · INTRODUCTION TO COMPUTER NETWORKS AND THE INTERNET · TERM 4

# Module 3

## Error Detection and Correction

[cosc264.up.railway.app](https://cosc264.up.railway.app)

## A quick recap of last week

There are two kinds of routing: **intra-AS routing**, inside a single autonomous system, and **inter-AS routing**, between them.

For **intra-AS routing** we looked at **RIP**, a distance-vector protocol, and **OSPF**, a link-state one.

For **inter-AS routing** there is effectively only one answer, **BGP**: **eBGP** runs between ASes and learns the AS-level path to prefixes in other ASes, and **iBGP** carries what was learned to every BGP speaker inside the AS.

We also looked at **scaling the IPv4 address space**, by two main mechanisms: **NAT** and **IPv6**.

MODULE 3 · CONCEPT 1 OF 3

# Error Detection

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1 · **Error Detection**

2 · CRC

3 · Error Correction

## MOTIVATION

# The journey of a packet

Consider sending a datagram from **host A** to **host B** across a network of links and routers.



Along the way — **thermal noise**, weak signals, interference, equipment crashes — the packet might arrive **corrupted**, arrive **out of order**, or be **dropped** entirely.

# Three categories of problem

| PROBLEM                                                                                                                                                                                               | APPROACH                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Corrupted packets                                                                                                                                                                                     | Error detection and correction <a href="#">this module</a> |
| <div><b>DEFINITION</b><br/>A packet is <b>corrupted</b> when it is delivered, but one or more of its bits have been <b>altered in transit</b>. Those altered bits are called <b>bit errors</b>.</div> |                                                            |
| Lost packets                                                                                                                                                                                          | ARQ protocols <a href="#">Module 4</a>                     |
| Out-of-order packets                                                                                                                                                                                  | ARQ protocols <a href="#">Module 4</a>                     |

# Types of bit errors



**Burst error** of length B — the span from the first to the last corrupted bit is B positions. The bits in between **may or may not be flipped**. Usually one physical event is behind it — a spike of interference, or a brief fade on a wireless link.

**Single-bit error** — a single isolated bit is altered.

# The error detection process

The sender runs the data through an **error-detecting code**, which produces a small number of **check bits**  $E$  — a compact fingerprint that depends on every bit of the original content.

## TRANSMITTER



# The error detection process

## RECEIVER

1. Receive and parse: 

|              |          |
|--------------|----------|
| <b>Data'</b> | <b>E</b> |
|--------------|----------|

 → 

|                        |
|------------------------|
| <b>Data'</b><br>k bits |
|------------------------|

 + 

|                            |
|----------------------------|
| <b>E</b><br>check bits (r) |
|----------------------------|
2. 

|              |
|--------------|
| <b>Data'</b> |
|--------------|

 → 

|                      |
|----------------------|
| recompute check bits |
|----------------------|

 → 

|                             |
|-----------------------------|
| <b>E'</b><br>check bits (r) |
|-----------------------------|
3. Compare **E** (received) with **E'** (recomputed)

**E = E'** → no error detected — accept the data.

**E ≠ E'** → error detected — discard or request retransmission.

- If nothing is corrupted, the check **always** passes.
- The reverse is **not** true. A check that passes does not guarantee nothing was corrupted — it only says **no error was detected**. Some rare corruption patterns still slip through.
- So a **good** error-detecting code should catch **as many error patterns as it can**: if an error occurred, it is detected with **high probability**.

# Parity check

The simplest **error-detecting** scheme: append a **parity bit** to each block of data.

**Even parity** — parity bit makes the total number of 1s (including the parity bit) even.

## TRANSMITTER

1. 

|                 |
|-----------------|
| 1 0 0 0 0 1 0 1 |
| data (3 ones)   |

 → 

|             |
|-------------|
| even parity |
|-------------|

 → 

|            |
|------------|
| 1          |
| parity bit |

 $3 + 1 = 4$  (even) ✓

2. Append: 

|                 |
|-----------------|
| 1 0 0 0 0 1 0 1 |
|-----------------|

|   |
|---|
| 1 |
|---|

 → transmit

## PARITY CHECK

# Parity check — detecting the error

### IN TRANSIT

Sent: 1 0 0 0 0 1 0 1 1 bit 4 flips → Received: 1 0 0 1 0 1 0 1 1 — the parity bit itself arrives unchanged

### RECEIVER

1. Parse: 1 0 0 1 0 1 0 1 data' (4 ones) + 1 received
2. 1 0 0 1 0 1 0 1 → even parity → 0 recomputed
3. Compare: **1** (received) ≠ **0** (recomputed) → **Error detected!**

**Odd parity** is the same idea with the opposite convention — the parity bit makes the total number of 1s **odd**.

## Parity check — can it be fooled?

Can you think of a scenario where the parity check **fails** — the data arrives corrupted, but the receiver sees nothing wrong?

Take the same even-parity example, and let **two** bits flip — one of them the parity bit itself.

### IN TRANSIT — 2 BITS FLIPPED

1 0 0 0 0 1 0 1 | 1 → 1 0 0 1 0 1 0 1 | 0 bit 4: 0→1, parity bit: 1→0

### RECEIVER

1. Parse: 

1 0 0 1 0 1 0 1  
data' (4 ones)

 + 

0  
received
2. 

1 0 0 1 0 1 0 1

 → 

even parity

 → 

0  
recomputed
3. Compare: 0 = 0 → **No error detected!**

## The limitation of a parity check

A parity check cannot detect a corruption pattern in which an **even number of bits** are flipped.

The reason is simple: adding an **even number** to a count never changes whether that count is odd or even. The recomputed parity bit comes out the same as the one received, and the check passes.

So we need a more **robust** error-detection scheme.

How? Intuitively, by spending **more check bits**.

That is the idea behind the **two-dimensional parity check**.

## 2-dimensional parity check

Arrange the data into a **matrix** of rows and columns.

For example, a 20-bit data block becomes the  $4 \times 5$  matrix on the right.

Append a **row parity bit**  $r_i$  to each row (even parity).

Append a **column parity bit**  $c_j$  to each column.

An **overall parity bit**  $p$  completes the matrix.

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 0 | 1 | 1 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 | 0 | 1 |
| 0 | 1 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 0 |

↑ column parity      row parity →

## How much can 2-D parity catch?

Flip **two bits in the same row** — the pattern that defeated the one-dimensional check.

Can this error be detected by the 2-D parity check?

The **row** check still passes: within that row, one flip cancels the other.

But each flip is alone in its **column**, so it flips that column's parity — the **column check catches the error**.

In general, every pattern of **one, two or three** flipped bits is detected.

Can you think of an error pattern that 2-D parity would miss?

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 1 | 0 | 1 | ✓ |
| 0 | 0 | 0 | 1 | 0 | 1 | ✓ |
| 0 | 1 | 0 | 0 | 0 | 1 | ✓ |
| 0 | 1 | 0 | 1 | 1 | 1 | ✓ |
| 0 | 0 | 0 | 1 | 1 | 0 |   |
| ✓ | x | x | ✓ | ✓ |   |   |

## How much can 2-D parity catch?

Flip **two bits in the same row** — the pattern that defeated the one-dimensional check.

Can this error be detected by the 2-D parity check?

The **row** check still passes: within that row, one flip cancels the other.

But each flip is alone in its **column**, so it flips that column's parity — the **column check catches the error**.

In general, every pattern of **one, two or three** flipped bits is detected.

Can you think of an error pattern that 2-D parity would miss?

Consider the four flipped bits on the right, at the corners of a **rectangle** — every check **passes**, and the error goes **undetected**.

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | 1 | ✓ |
| 0 | 0 | 0 | 1 | 0 | 1 | ✓ |
| 0 | 1 | 0 | 0 | 0 | 1 | ✓ |
| 0 | 1 | 0 | 1 | 1 | 1 | ✓ |
| 0 | 0 | 0 | 1 | 1 | 0 |   |
| ✓ | ✓ | ✓ | ✓ | ✓ |   |   |

## 2-D PARITY

# Correcting a single-bit error

2-D parity can not only **detect** an error — if just a single bit was flipped, it can also **correct** it.

The matrix on the right is as transmitted.

During transmission, a single bit is **flipped**.

The receiver's parity check **fails for row 2** and **column 2**.

The failing row and column **intersect** at the corrupted bit — flip it to **correct** the error.

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 1 | 0 | 1 | ✓ |
| 0 | 1 | 1 | 1 | 0 | 1 | ✗ |
| 0 | 1 | 0 | 0 | 0 | 1 | ✓ |
| 0 | 1 | 0 | 1 | 1 | 1 | ✓ |
| 0 | 0 | 0 | 1 | 1 | 0 |   |
| ✓ | ✗ | ✓ | ✓ | ✓ |   |   |

# Internet checksum

Parity is the simplest form of the idea. The **Internet checksum** is the scheme real protocols actually carry.

The **Internet checksum** is a specific **error-detecting code** used in **IPv4**, **TCP**, and **UDP** (discussed later) to let the receiver detect bit errors introduced during transmission.

IPv4 HEADER — 32 bits

|                           |          |                |                 |                 |
|---------------------------|----------|----------------|-----------------|-----------------|
| Version                   | IHL      | Diff. Services | Total Length    |                 |
| Identification            |          |                | Flags           | Fragment Offset |
| TTL                       | Protocol |                | Header Checksum |                 |
| Source Address            |          |                |                 |                 |
| Destination Address       |          |                |                 |                 |
| Options (0 or more words) |          |                |                 |                 |

TCP HEADER — 32 bits

|                                  |     |       |                  |                |  |
|----------------------------------|-----|-------|------------------|----------------|--|
| Source Port                      |     |       | Destination Port |                |  |
| Sequence Number                  |     |       |                  |                |  |
| Acknowledgement Number           |     |       |                  |                |  |
| Data Offset                      | Rsv | Flags |                  | Window Size    |  |
| Checksum                         |     |       |                  | Urgent Pointer |  |
| Options (0 or more 32-bit words) |     |       |                  |                |  |
| Data (optional)                  |     |       |                  |                |  |

# One's complement arithmetic

The checksum uses two operations:

## One's complement addition

1. Add the two numbers as unsigned integers.
2. If there is a carry out of the leftmost bit, add 1 to the sum (*end-around carry*).

ex1. (no carry)

```
A2B3
+ 1D4C
-----
BFFF
```

ex2. (carry out)

```
A2B3
+ 6F8E
-----
carry → 11241
+   1 ← end-around
-----
1242
```

## One's complement operation

Replace every 0 bit with 1 and every 1 bit with 0.

1111 → ones complement → 0000

1001 → ones complement → 0110

Adding a value and its one's complement always gives all 1s:

1001 + 0110 = 1111

# Sender computation

- 1. Take the **IP header**.
- 2. Split it into **16-bit words**.
- 3. Compute their **one's-complement sum**.
- 4. Take the **one's complement** of that sum — that is the **checksum**.

IPv4 header rows are 32 bits, so each can be written as **8 hex digits**:

F2345678

F234 5678 32 bits → two 16-bit words

|                       | HEX   | BINARY                |
|-----------------------|-------|-----------------------|
| Word 1                | F234  | 1111 0010 0011 0100   |
| Word 2                | 5678  | 0101 0110 0111 1000   |
| Add                   | 148AC | 1 0100 1000 1010 1100 |
| Wrap carry            | + 1   | + 1                   |
| One's-complement sum  | 48AD  | 0100 1000 1010 1101   |
| Checksum = complement | B752  | 1011 0111 0101 0010   |

# Receiver verification

- 1. Take the received header — the checksum field now holds **B752**.
- 2. Compute the **one's-complement sum** of the header words.
- 3. Add the **checksum** to that sum.
- 4. The checksum is the **one's complement** of that sum, so adding them must give **all 1 bits** — check that the total is **FFFF**.

F234 5678 **B752**

|                      | HEX    | BINARY                |
|----------------------|--------|-----------------------|
| Word 1               | F234   | 1111 0010 0011 0100   |
| Word 2               | 5678   | 0101 0110 0111 1000   |
| Add words 1–2        | 148AC  | 1 0100 1000 1010 1100 |
| Wrap carry           | + 1    | + 1                   |
| One's-complement sum | 48AD   | 0100 1000 1010 1101   |
| Add checksum         | + B752 | + 1011 0111 0101 0010 |
| Total                | FFFF   | 1111 1111 1111 1111   |

# Efficiency and limitations

The primary reason for its adoption is **efficiency**:

- Most of these protocols are implemented in software
- The Internet checksum involves simple operations → very little computational overhead

But it also has a **limitation**:

- It is much less effective at detecting errors than the **cyclic redundancy check (CRC)**, which we look at next

MODULE 3 · CONCEPT 2 OF 3

# Cyclic Redundancy Check

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1 · Error Detection

2 · **CRC**

3 · Error Correction

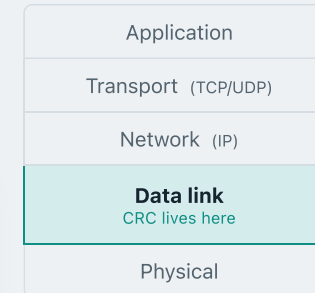
# Cyclic Redundancy Check (CRC)

The CRC is one of the **most common and most powerful** error-detecting codes in use.

It lives in the **data link layer**. Every frame you send over Ethernet or Wi-Fi carries a CRC.

## The intuition

- This time we read the message as a **number** — one big integer, rather than a string of bits.
- Sender and receiver agree on a fixed **divisor** in advance.
- The sender appends a few **check bits**, chosen so that the whole thing — read as one number — is **exactly divisible** by that divisor.



## CRC — notation

**$D$**  — the  $k$ -bit block of data to send

**$P$**  — the **divisor**, a fixed pattern of  $r+1$  bits that sender and receiver agree on in advance

**$F$**  — the **frame check sequence (FCS)** we must compute,  $r$  bits long

**$T$**  — the frame actually transmitted,  $k+r$  bits:  $D$  followed by  $F$

Some texts write  $F$ 's length as  $n-k$ , where  $n = k+r$  is the whole frame.



## How to choose the pattern P?

Not every pattern will do — two **clearly bad** choices:

- **P = 0000...0** — you cannot divide by zero at all.
- **Highest-order bit of P = 0** — the leading zero contributes nothing to the division, so P is effectively **one bit shorter**. And since the FCS is always one bit shorter than P, the **effective FCS is one bit shorter** too — fewer check bits than we intended.

Intuitively, fewer check bits means weaker error detection — not what we want.

The exact bit pattern depends on the **type of errors** expected — but in every case:

### REQUIRED

The **highest-order bit** = 1

### REQUIRED

The **lowest-order bit** = 1

Take this one on trust for now — we will work an example of why later.

Together these two buy the **burst-error guarantee** — we will see why shortly.

## Modulo-2 Arithmetic

- Addition and subtraction are both performed with **XOR ( $\oplus$ )** — so they are the **same operation**.

### ADDITION (XOR)

$$\begin{array}{r} 1111 \\ \oplus 1010 \\ \hline 0101 \end{array}$$

### SUBTRACTION (XOR)

$$\begin{array}{r} 1111 \\ \oplus 1010 \\ \hline 0101 \end{array}$$

- Multiplication is the usual shift-and-add — only the additions inside it become XORs.

### MULTIPLICATION

$$\begin{array}{r} 11001 \\ \times 11 \\ \hline 11001 \\ \oplus 110010 \\ \hline 101011 \end{array}$$

# How to compute the FCS

## GIVEN

$D$  **1010001101**  $k = 10$  bits

$P$  **110101**  $r+1 = 6$  bits

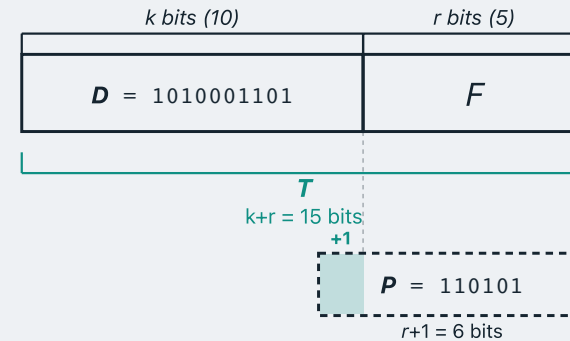
## TO FIND

$F$ , the **FCS** — one bit shorter than  $P$ , so  $r = 5$  bits — such that:

1. appending  $F$  to  $D$  gives  $T$ , of  $k+r = 10 + 5 = 15$  bits;
2. as integers,  $T = D \times 2^r + F$ ;
3. and  $T$  is **exactly divisible by  $P$** .

So what should  $F$  be?

$$F = \text{remainder of } (D \times 2^r) / P$$



Plugging in the numbers:

$$\begin{aligned} F &= \text{rem. of } (1010001101 \times 2^5) / 110101 \\ &= \text{rem. of } 101000110100000 / 110101 \end{aligned}$$

## An Example — Computing FCS

```

          1101010110    quotient (discarded)
          -----
110101 ) 101000110100000    D × 25
          110101000000000    ⊕ P
          -----
          11101110100000
          110101000000000    ⊕ P shifted
          -----
          111010100000
          : four more steps :
          -----
          01110    remainder
  
```

We need the remainder of  $101000110100000 \div 110101$ , and we get it by **long division** — the same procedure you learned at school.

The only difference: at each step the subtraction is an **XOR**, because we are working in **modulo-2 arithmetic**.

**$F = 01110$**  (the 5-bit FCS)

Transmitted frame  $T$ :

1010001101**01110**

## The receiver's check

```

          1101010110 quotient (discarded)
          -----
110101 ) 101000110101110 received T
        1101010000000000 ⊕ P
        -----
          11101110101110
          1101010000000000 ⊕ P shifted
          -----
            111010101110
              : four more steps :
              -----
                00000 remainder
  
```

The receiver gets the transmitted frame  $T$  and knows the divisor  $P$ . To check whether the frame is corrupted, it simply checks whether  $T$  is **divisible by  $P$**  — again, by long division.

**Remainder = 0**

In this case  $T$  is divisible by  $P$ , so the receiver declares that **no error is detected**.

# Polynomial representation

The divisor  $P$  is also called the **generator** — it is what generates the FCS.

A bit string can also be written as a **polynomial**, and generators are almost always written that way.

Two reasons to use the polynomial representation:

- Every CRC **standard** publishes its generator in this form.
- It is the form in which you can **analyse** which errors a generator will catch.

# Bit strings as polynomials

Each bit becomes one term: the **power** of  $x$  gives the bit's **position**, the **coefficient** gives its **value**. Only the 1-bits survive.

| $x^9$ | $x^8$ | $x^7$ | $x^6$ | $x^5$ | $x^4$ | $x^3$ | $x^2$ | $x^1$ | $x^0$ |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1     | 0     | 1     | 0     | 0     | 0     | 1     | 1     | 0     | 1     |

 $\rightarrow$ 

$x^9 + x^7 + x^3 + x^2 + 1$

binary  $\rightarrow$  polynomial

Going back, write a 1 wherever a power appears and a 0 wherever it does not — here  $x^6$ ,  $x^3$ ,  $x^2$  and  $x^1$  are all missing.

$x^7 + x^5 + x^4 + 1$

 $\rightarrow$ 

| $x^7$ | $x^6$ | $x^5$ | $x^4$ | $x^3$ | $x^2$ | $x^1$ | $x^0$ |
|-------|-------|-------|-------|-------|-------|-------|-------|
| 1     | 0     | 1     | 1     | 0     | 0     | 0     | 1     |

polynomial  $\rightarrow$  binary

## The same FCS, two representations

### BIT STRINGS

$D$  1010001101

$P$  110101

$F$  = remainder of  
101000110100000  $\div$  110101

### POLYNOMIALS

$D(x) = x^9 + x^7 + x^3 + x^2 + 1$

$P(x) = x^5 + x^4 + x^2 + 1$

$F(x)$  = remainder of  
?

How do we write "append five zeros to  $D$ , then divide by  $P$ " in polynomial form?

# The same FCS, two representations

## BIT STRINGS

$D$  1010001101

$P$  110101

$F$  = remainder of  
101000110100000  $\div$  110101

## POLYNOMIALS

$D(x) = x^9 + x^7 + x^3 + x^2 + 1$

$P(x) = x^5 + x^4 + x^2 + 1$

$F(x)$  = remainder of  
 $D(x) \cdot x^5 \div P(x)$

How do we write “append five zeros to  $D$ , then divide by  $P$ ” in polynomial form?

Multiply  $D(x)$  by  $x^5$  and divide by  $P(x)$  — it does exactly the same thing. Worth checking for yourself.

**You do not need to learn polynomial division.** If it is unfamiliar, work in bit strings and convert the remainder at the end.

$F$  01110

$F(x) = x^3 + x^2 + x$

$T$  101000110101110

$T(x) = x^{14} + x^{12} + x^8 + x^7 + x^5 + x^3 + x^2 + x$

# Why is this representation useful?

Optional material, for anyone curious about *why* the polynomial form is used. Not examined.

It all comes down to one question: **how do we choose a good generator  $P(x)$ ?**

## OBSERVATION

A corrupted frame can always be written as  $T(x) + E(x)$ , where the **error polynomial**  $E(x)$  has a 1 in every flipped position.

CRC builds  $T(x)$  so that  $P(x)$  divides it **exactly**. So the receiver's check is simply: is  $T(x) + E(x)$  divisible by  $P(x)$ ?

Therefore the error slips through **exactly when  $P(x)$  divides  $E(x)$** .

So a well-chosen  $P(x)$  is one that **cannot divide** the error patterns we expect to see — which is exactly what makes it catch most of them.

Recall that we required  $P$ 's **highest-order bit** to be 1. Here is why.

## Example: a short burst error

Our generator is  $P(x) = x^5 + x^4 + x^2 + 1$ , of **degree 5**.

The **degree** of a polynomial is its highest power — equivalently, the position of its leading 1 bit.

|                 | $x^{14}$ | $x^{13}$ | $x^{12}$ | $x^{11}$ | $x^{10}$ | $x^9$ | $x^8$ | $x^7$ | $x^6$ | $x^5$ | $x^4$ | $x^3$ | $x^2$ | $x^1$ | $x^0$ |
|-----------------|----------|----------|----------|----------|----------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| <b>Sent T</b>   | 1        | 0        | 1        | 0        | 0        | 0     | 1     | 1     | 0     | 1     | 0     | 1     | 1     | 1     | 0     |
| <b>Received</b> | 1        | 0        | 1        | 0        | 0        | 0     | 1     | 1     | 0     | 1     | 0     | 1     | 0     | 0     | 1     |
| <b>Error E</b>  | 0        | 0        | 0        | 0        | 0        | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 1     |

E has a 1 exactly where a bit was flipped — here a burst of three

The flipped bits are the last three, so the error polynomial is  $E(x) = x^2 + x + 1$  — of **degree 2**.

## Example: a short burst error

Our generator is  $P(x) = x^5 + x^4 + x^2 + 1$ , of **degree 5**.

The **degree** of a polynomial is its highest power — equivalently, the position of its leading 1 bit.

|                 | $x^{14}$ | $x^{13}$ | $x^{12}$ | $x^{11}$ | $x^{10}$ | $x^9$ | $x^8$ | $x^7$ | $x^6$ | $x^5$ | $x^4$ | $x^3$ | $x^2$ | $x^1$ | $x^0$ |
|-----------------|----------|----------|----------|----------|----------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| <b>Sent T</b>   | 1        | 0        | 1        | 0        | 0        | 0     | 1     | 1     | 0     | 1     | 0     | 1     | 1     | 1     | 0     |
| <b>Received</b> | 1        | 0        | 1        | 0        | 0        | 0     | 1     | 1     | 0     | 1     | 0     | 1     | 0     | 0     | 1     |
| <b>Error E</b>  | 0        | 0        | 0        | 0        | 0        | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 1     |

E has a 1 exactly where a bit was flipped — here a burst of three

A degree-5 polynomial cannot divide a non-zero one of degree 2, so the remainder cannot be zero: **this error is caught.**

## Example: a short burst error

Our generator is  $P(x) = x^5 + x^4 + x^2 + 1$ , of **degree 5**.

The **degree** of a polynomial is its highest power — equivalently, the position of its leading 1 bit.

|                 | $x^{14}$ | $x^{13}$ | $x^{12}$ | $x^{11}$ | $x^{10}$ | $x^9$ | $x^8$ | $x^7$ | $x^6$ | $x^5$ | $x^4$ | $x^3$ | $x^2$ | $x^1$ | $x^0$ |
|-----------------|----------|----------|----------|----------|----------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| <b>Sent T</b>   | 1        | 0        | 1        | 0        | 0        | 0     | 1     | 1     | 0     | 1     | 0     | 1     | 1     | 1     | 0     |
| <b>Received</b> | 1        | 0        | 1        | 0        | 0        | 0     | 1     | 1     | 0     | 1     | 0     | 1     | 0     | 0     | 1     |
| <b>Error E</b>  | 0        | 0        | 0        | 0        | 0        | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 1     |

E has a 1 exactly where a bit was flipped — here a burst of three

This is why we required the **highest-order bit to be 1** — it is what holds the degree at  $r$ , and the degree is what sets how long a burst can be and still be caught.

## Example: a short burst error

Our generator is  $P(x) = x^5 + x^4 + x^2 + 1$ , of **degree 5**.

The **degree** of a polynomial is its highest power — equivalently, the position of its leading 1 bit.

|                 | $x^{14}$ | $x^{13}$ | $x^{12}$ | $x^{11}$ | $x^{10}$ | $x^9$ | $x^8$ | $x^7$ | $x^6$ | $x^5$ | $x^4$ | $x^3$ | $x^2$ | $x^1$ | $x^0$ |
|-----------------|----------|----------|----------|----------|----------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| <b>Sent T</b>   | 1        | 0        | 1        | 0        | 0        | 0     | 1     | 1     | 0     | 1     | 0     | 1     | 1     | 1     | 0     |
| <b>Received</b> | 1        | 0        | 1        | 0        | 0        | 0     | 1     | 1     | 0     | 1     | 0     | 1     | 0     | 0     | 1     |
| <b>Error E</b>  | 0        | 0        | 0        | 0        | 0        | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 1     |

E has a 1 exactly where a bit was flipped — here a burst of three

We also required the **lowest-order bit** to be 1. Why? That one is trickier.

## Example: a short burst error

Our generator is  $P(x) = x^5 + x^4 + x^2 + 1$ , of **degree 5**.

The **degree** of a polynomial is its highest power — equivalently, the position of its leading 1 bit.

|                 | $x^{14}$ | $x^{13}$ | $x^{12}$ | $x^{11}$ | $x^{10}$ | $x^9$ | $x^8$ | $x^7$ | $x^6$ | $x^5$ | $x^4$ | $x^3$ | $x^2$ | $x^1$ | $x^0$ |
|-----------------|----------|----------|----------|----------|----------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| <b>Sent T</b>   | 1        | 0        | 1        | 0        | 0        | 0     | 1     | 1     | 0     | 1     | 0     | 1     | 1     | 1     | 0     |
| <b>Received</b> | 1        | 0        | 1        | 0        | 0        | 0     | 1     | 0     | 1     | 1     | 1     | 1     | 1     | 1     | 0     |
| <b>Error E</b>  | 0        | 0        | 0        | 0        | 0        | 0     | 0     | 1     | 1     | 0     | 1     | 0     | 0     | 0     | 0     |

E has a 1 exactly where a bit was flipped — here a burst of four, away from the end of the frame

Suppose the burst is not at the end of the frame but in the **middle** — still only **four** bits wide. Sitting away from the end, it factors as a short pattern **shifted along**:  $E(x) = x^4 \cdot (x^3 + x^2 + 1)$ .

## Example: a short burst error

Our generator is  $P(x) = x^5 + x^4 + x^2 + 1$ , of **degree 5**.

The **degree** of a polynomial is its highest power — equivalently, the position of its leading 1 bit.

|                 | $x^{14}$ | $x^{13}$ | $x^{12}$ | $x^{11}$ | $x^{10}$ | $x^9$ | $x^8$ | $x^7$ | $x^6$ | $x^5$ | $x^4$ | $x^3$ | $x^2$ | $x^1$ | $x^0$ |
|-----------------|----------|----------|----------|----------|----------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| <b>Sent T</b>   | 1        | 0        | 1        | 0        | 0        | 0     | 1     | 1     | 0     | 1     | 0     | 1     | 1     | 1     | 0     |
| <b>Received</b> | 1        | 0        | 1        | 0        | 0        | 0     | 1     | 0     | 1     | 1     | 1     | 1     | 1     | 1     | 0     |
| <b>Error E</b>  | 0        | 0        | 0        | 0        | 0        | 0     | 0     | 1     | 1     | 0     | 1     | 0     | 0     | 0     | 0     |

E has a 1 exactly where a bit was flipped — here a burst of four, away from the end of the frame

Now suppose  $P$ 's last bit were **0** instead of 1 — the constant term goes, and  $P(x) = x^5 + x^4 + x^2 + 1$  becomes  $x^5 + x^4 + x^2$ , which factors as  $x^2 \cdot (x^3 + x^2 + 1)$ . And since  $E(x) = x^4 \cdot (x^3 + x^2 + 1)$ ,  $E$  is exactly  $x^2 \cdot P$ : it divides with no remainder, and a burst of only four **slips through**.

## Example: a short burst error

Our generator is  $P(x) = x^5 + x^4 + x^2 + 1$ , of **degree 5**.

The **degree** of a polynomial is its highest power — equivalently, the position of its leading 1 bit.

|                 | $x^{14}$ | $x^{13}$ | $x^{12}$ | $x^{11}$ | $x^{10}$ | $x^9$ | $x^8$ | $x^7$ | $x^6$ | $x^5$ | $x^4$ | $x^3$ | $x^2$ | $x^1$ | $x^0$ |
|-----------------|----------|----------|----------|----------|----------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| <b>Sent T</b>   | 1        | 0        | 1        | 0        | 0        | 0     | 1     | 1     | 0     | 1     | 0     | 1     | 1     | 1     | 0     |
| <b>Received</b> | 1        | 0        | 1        | 0        | 0        | 0     | 1     | 0     | 1     | 1     | 1     | 1     | 1     | 1     | 0     |
| <b>Error E</b>  | 0        | 0        | 0        | 0        | 0        | 0     | 0     | 1     | 1     | 0     | 1     | 0     | 0     | 0     | 0     |

E has a 1 exactly where a bit was flipped — here a burst of four, away from the end of the frame

Our  $P$  ends in 1, so it carries **no factor of  $x$** : it cannot divide the  $x^4$ , and  $x^3 + x^2 + 1$  is too small to divide.  
**Caught.** This is why the **lowest-order bit** must be 1 as well.

## Some Standard Polynomials

These trade-offs have already been worked out. **Always use a standard generator — never invent your own.**

Four versions of  $P(X)$  are widely used:

- **CRC-12**: transmissions of streams of 6-bit characters
- **CRC-16 / CRC-ANSI**: various communication protocols, including serial standards like Modbus
- **CRC-CCITT**: telecommunications applications
- **CRC-32 / IEEE-802**: network protocols — Ethernet, ZIP, PNG, and other file transfer protocols

|                  |                                                                                                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>CRC-12</b>    | $= X^{12} + X^{11} + X^3 + X^2 + X + 1$                                                                              |
| <b>CRC-ANSI</b>  | $= X^{16} + X^{15} + X^2 + 1$                                                                                        |
| <b>CRC-CCITT</b> | $= X^{16} + X^{12} + X^5 + 1$                                                                                        |
| <b>IEEE-802</b>  | $= X^{32} + X^{26} + X^{23} + X^{22} + X^{16} + X^{12} + X^{11} + X^{10} + X^8$<br>$+ X^7 + X^5 + X^4 + X^2 + X + 1$ |

## RECAP

# Error detection — what we covered

- An **error-detecting code** appends check bits so the receiver can test whether the data arrived intact.
- A **parity check** adds a single bit, and misses any corruption that flips an **even number** of bits.
- **Two-dimensional parity** checks rows and columns. It catches any one, two or three flips and can even **correct** a single-bit error — but not four at the corners of a rectangle.
- The **Internet checksum** is stronger than a parity check, and **very cheap to compute** — which is why **IP, TCP and UDP** all use it.
- The **CRC** appends an FCS that makes the frame divide exactly by an agreed generator — the strongest of them all.

MODULE 3 · CONCEPT 3 OF 3

# Error Correction

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1 · Error Detection

2 · CRC

3 · Error Correction

# Forward Error Correction (FEC)

→ Typically, **retransmission** is needed when bit errors are detected.

→ Not good enough for **wireless applications**:

- High **bit error rate (BER)** on a wireless link
- Long **propagation delay** on satellite links

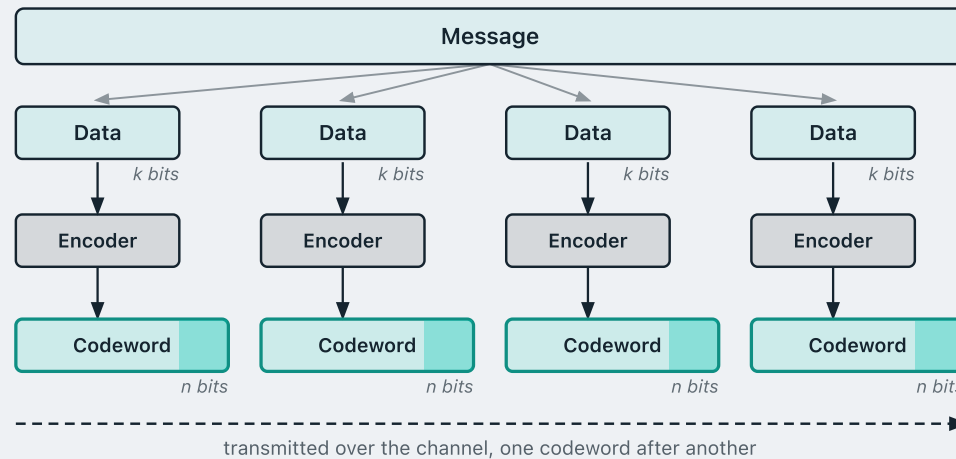
→ Need to **correct errors** on the basis of the bits received — without retransmission.

# The FEC process — sender

The message is divided into blocks of  $k$  bits — each one a **data block**.

The **encoder** is applied to each data block **independently**, turning every  $k$ -bit block into an  $n$ -bit **codeword** — the extra  $n-k$  bits are the redundancy.

The codewords are then transmitted, one after another.



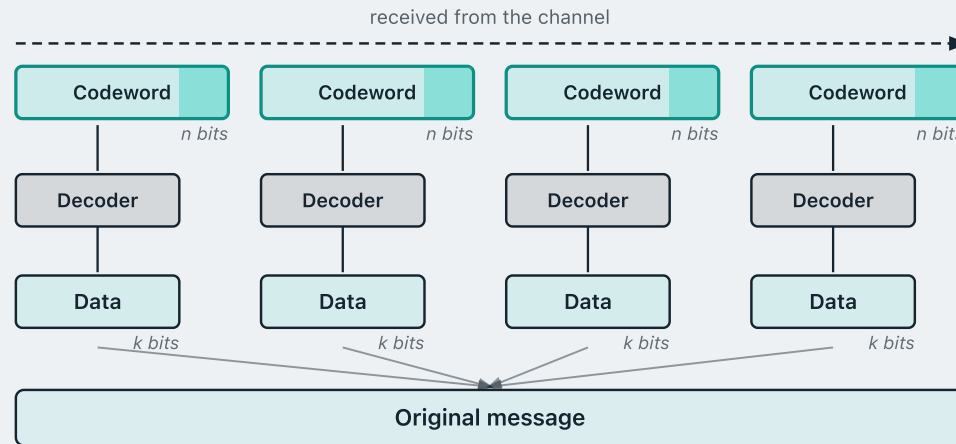
## The FEC process — receiver

The receiver splits the incoming stream back into  $n$ -bit codewords — some of which may have been corrupted in transit.

The **decoder** is applied to each codeword independently.

Each decode recovers the original  $k$ -bit data block, **correcting** any errors the code is strong enough to fix.

The recovered blocks are reassembled into the original message.



## Block Code Principles — Hamming Distance

The **Hamming distance**  $d(v_1, v_2)$  between two  $n$ -bit binary sequences is *the number of bit positions in which  $v_1$  and  $v_2$  disagree*.

Example:

$$\begin{array}{rcccccc} v_1 = & 0 & 1 & 1 & 0 & 1 & 1 \\ & \downarrow & \cdot & \downarrow & \cdot & \downarrow & \cdot \\ v_2 = & 1 & 1 & 0 & 0 & 0 & 1 \end{array}$$

3 positions differ  $\rightarrow d(v_1, v_2) = 3$

## How the encoder and decoder work

Constructing the **encoder** is the most important part — it is what determines how many errors the code can **detect** and **correct**. We will not go into those details here.

You can always think of an encoder as a **map**: a table assigning each  $k$ -bit data block one  $n$ -bit codeword. That is all we need in this lecture.

But whatever the code, the **decoder** always works in the same way.

for example, with  $k = 2$  and  $n = 5$ :

| Data block | Codeword |
|------------|----------|
| 00         | 00000    |
| 01         | 00111    |
| 10         | 11001    |
| 11         | 11110    |

# Minimum Hamming distance decoding

Given the received  $n$ -bit pattern, the decoder asks:

## 1. Is it one of the valid codewords?

If so, **no error is detected** — output the data block the encoder assigned to it.

## 2. If not, an error has occurred — that much we know immediately.

Then find the valid codeword **closest to it** in Hamming distance, and output *that* codeword's data block. This is **minimum-distance decoding**.

Step 2 works because **fewer bit flips are more likely than many** — the nearest valid codeword is the most plausible thing the sender actually sent.

## Minimum-distance decoding — an example

Our **encoder**, with  $k = 2$  and  $n = 5$ :

| Data block | Codeword |
|------------|----------|
| 00         | 00000    |
| 01         | 00111    |
| 10         | 11001    |
| 11         | 11110    |

**Assumption:** the channel is lossy, but **at most one bit** per codeword is flipped.

The receiver gets **00100** — which is **not** one of the four valid codewords, so something was corrupted.

Which valid codeword is closest?

| Received     | Hamming distance | Valid codeword | Data block |
|--------------|------------------|----------------|------------|
| <b>00100</b> | <b>1</b>         | <b>00000</b>   | <b>00</b>  |
| 00100        | 2                | 00111          | 01         |
| 00100        | 4                | 11001          | 10         |
| 00100        | 3                | 11110          | 11         |

The minimum distance is **1**, to 00000, so the decoder outputs data block **00**. Under our assumption of at most one flip, that **must** be what the transmitter sent — the error is corrected.

# When can we actually correct?

Minimum-distance decoding only works if the closest valid codeword is **unique**. It need not be.

The same encoder as before:

| Data block | Codeword |
|------------|----------|
| 00         | 00000    |
| 01         | 00111    |
| 10         | 11001    |
| 11         | 11110    |

**New assumption:** this channel is noisier — **up to two bits** per codeword may be flipped.

The receiver gets **01010**.

| Received     | Hamming distance | Valid codeword | Data block |
|--------------|------------------|----------------|------------|
| <b>01010</b> | <b>2</b>         | <b>00000</b>   | <b>00</b>  |
| 01010        | 3                | 00111          | 01         |
| 01010        | 3                | 11001          | 10         |
| <b>01010</b> | <b>2</b>         | <b>11110</b>   | <b>11</b>  |

**Two** codewords sit at distance 2, and two flips are now allowed. We know an error occurred — but 00 and 11 are equally plausible, so the error can be **detected but not corrected**.

## The full picture for this code

With  $n = 5$  there are  $2^5 = 32$  possible received patterns:

00000 00001 00010 ... 00111 ... 11001 ... 11110 11111

Only **four** of them are valid codewords, so  $32 - 4 = 28$  are invalid.

- **20 of the 28** are at Hamming distance 1 from a valid codeword.
- Each is at distance 1 from **exactly one** codeword, never two — so its nearest codeword is **unique**.
- Therefore all 20 can be **corrected**. That was our first example.

01010 01011 01100 01101 10010 10011 10100 10101

- The remaining **8** lie at distance 2 from **two different** codewords, so the nearest is **not unique**.
- So they can only be **detected** — our second example.

So this code can **detect** any error of up to two bits, but only **single-bit errors** can be **corrected**.

# What determines the ability to correct?

For a **general** code, how many errors can it correct, and how many can it detect?

It comes down to a single number: the **minimum Hamming distance**  $d$  of the code — the smallest distance between **any two** valid codewords.

Our code has four valid codewords:

00000

00111

11001

11110

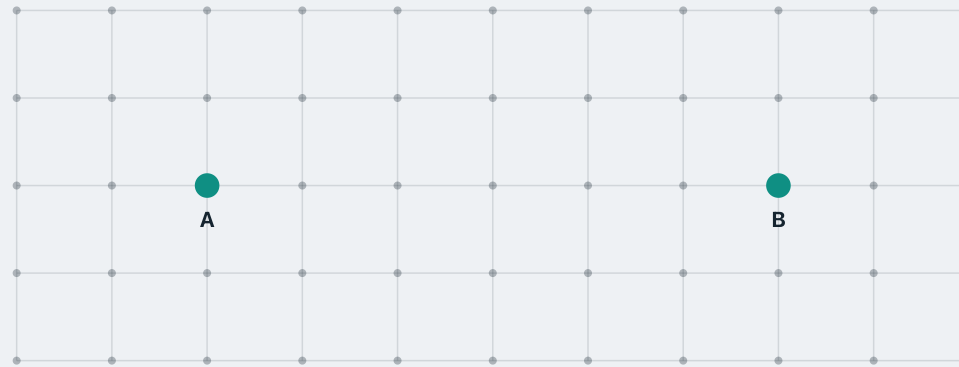
| Pair of codewords |       | Distance |
|-------------------|-------|----------|
| 00000             | 00111 | 3        |
| 00000             | 11001 | 3        |
| 00000             | 11110 | 4        |
| 00111             | 11001 | 4        |
| 00111             | 11110 | 3        |
| 11001             | 11110 | 3        |

Every pair, six in total.

Reading off the smallest entry:  **$d = 3$** .

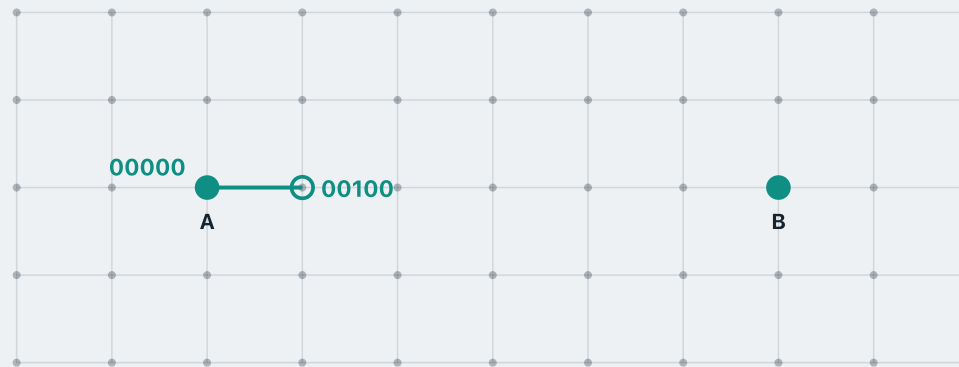
# Walking away from a codeword

Picture the whole space of bit strings as a **grid**: every node is one specific bit string, and any two **adjacent** nodes differ in **exactly one bit** — Hamming distance 1.



## Walking away from a codeword

For example, if **A** is 00000, then one step away is a string like 00100 — they differ in **exactly one bit**, so their Hamming distance is **1**.



# Walking away from a codeword

Step **up** instead and we reach 10000 — a different bit flipped, but still **one step** from 00000.



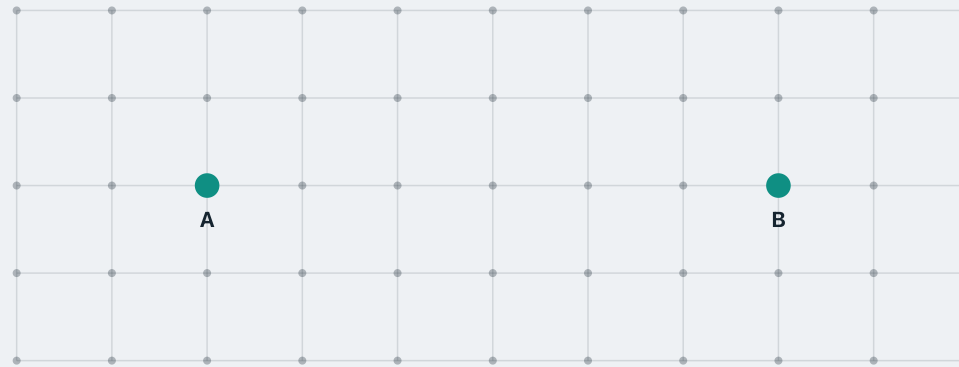
# Walking away from a codeword

Notice that 10000 and 00100 are **two steps** apart — two edges to travel from one to the other.



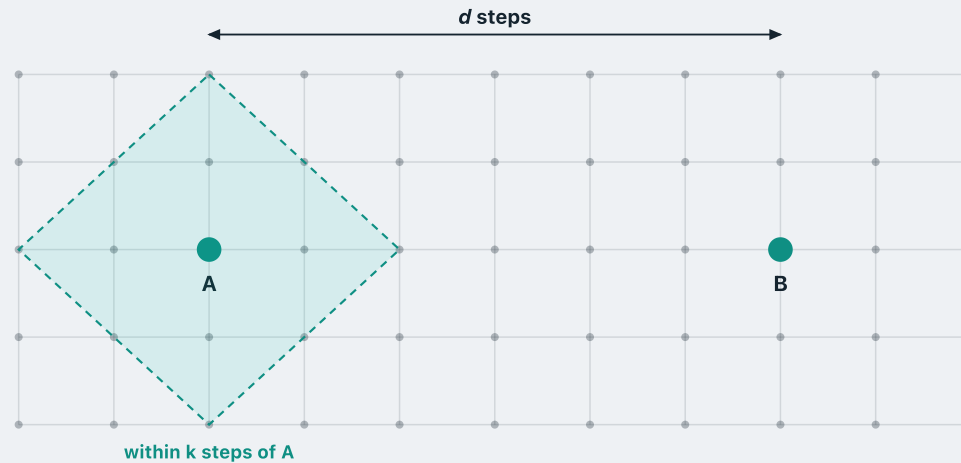
# Walking away from a codeword

Now suppose we use a code whose **minimum Hamming distance** is  $d$ .



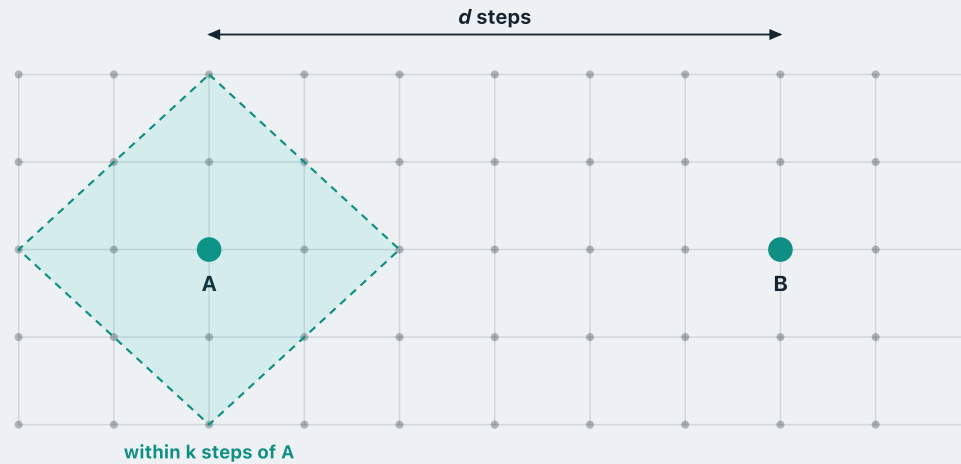
# Walking away from a codeword

Send **A** over a channel that flips at most  **$k$**  bits. Each flip is one step to a neighbour, so the received pattern lands **somewhere within  $k$  steps of A**.



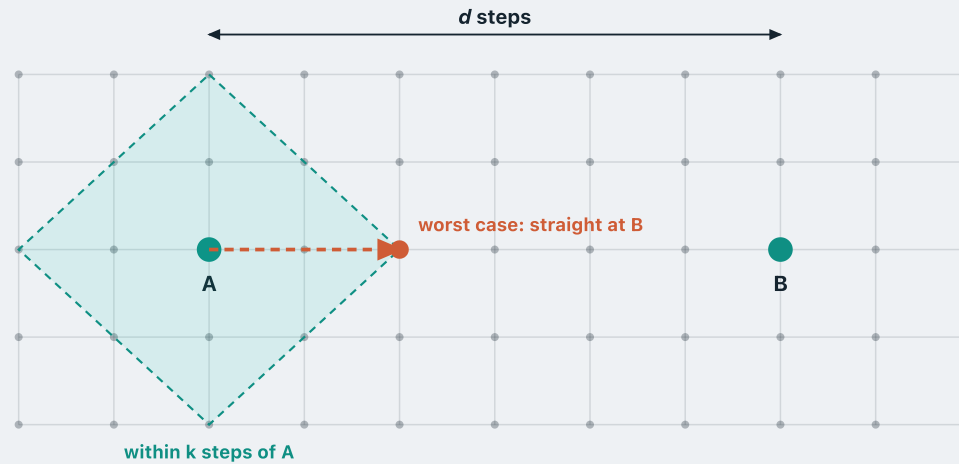
# Walking away from a codeword

The further it drifts from A, the **less it looks like A** — and the more it may start to resemble some other codeword.



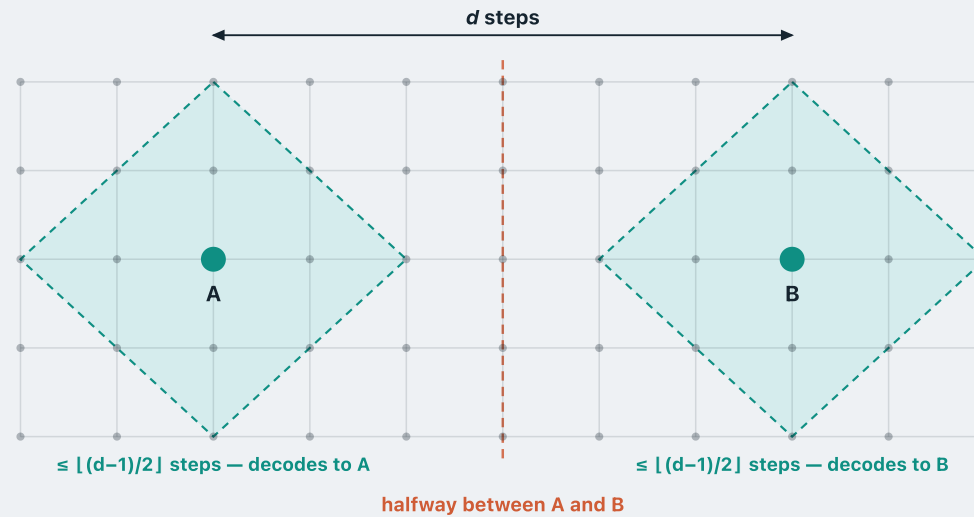
# Walking away from a codeword

The **worst case** is when every flip takes us straight towards the **nearest** codeword B, so the pattern looks more and more like B with each step.



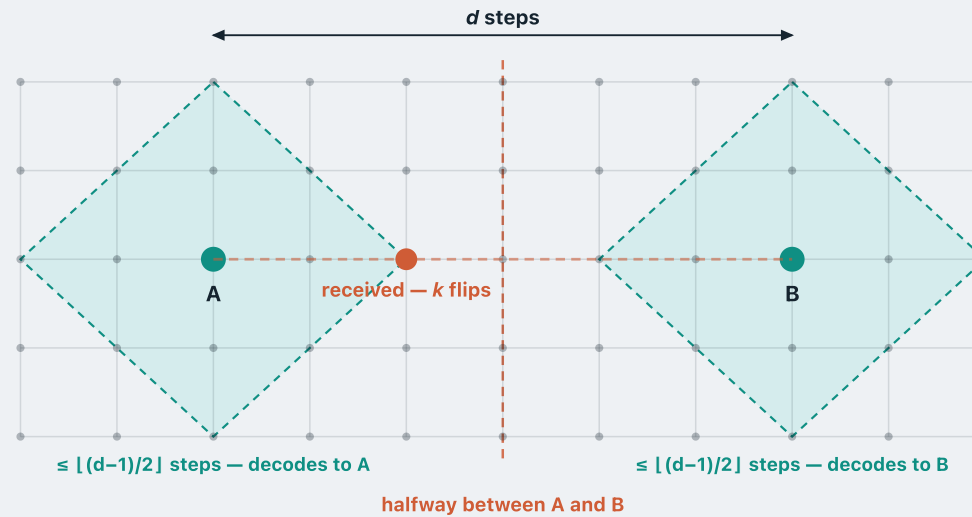
# When can the error still be corrected?

Suppose the error flips  $k$  bits, with  $k \leq \lfloor (d-1)/2 \rfloor$ . Then the received pattern stays inside **A's region** and never crosses the **halfway** line — even in the worst case.



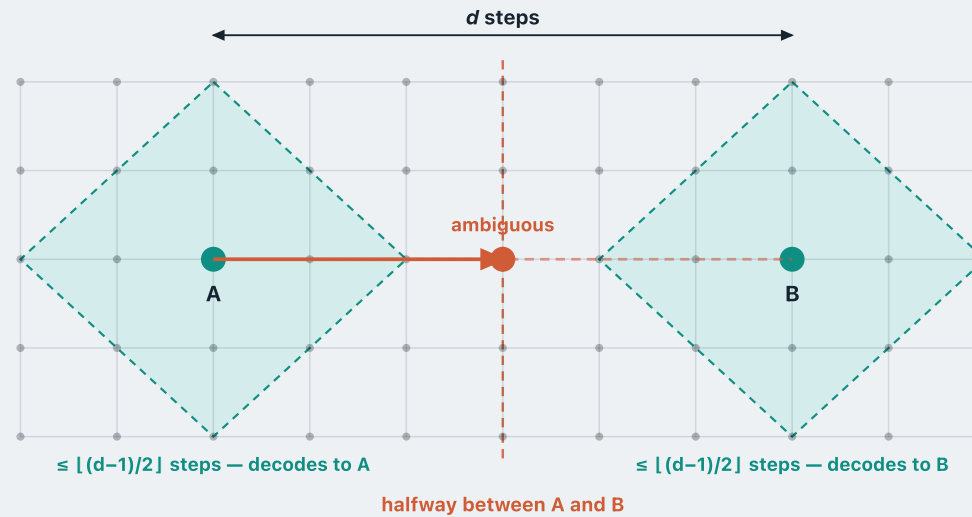
# When can the error still be corrected?

So **A** is strictly the nearest codeword, and minimum-distance decoding walks us back to it: a  $k$ -bit error is **corrected**.



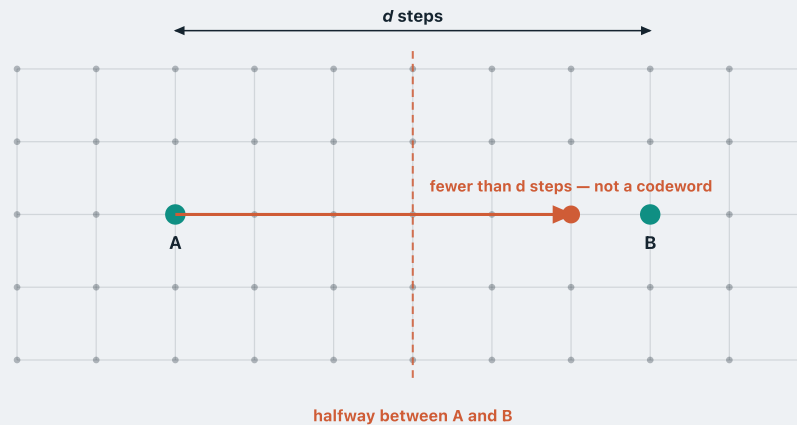
# When can the error still be corrected?

Now suppose  $k = d/2$ . The pattern lands exactly on the **halfway** line, **equidistant** from A and B — it could have come from either, so **correction is no longer possible**.



## Past halfway, we can still detect

- Suppose the channel flips **more than  $\lfloor (d-1)/2 \rfloor$**  bits, so correction is out of reach.
- As long as it flips **fewer than  $d$**  bits, we cannot reach B — B is a full  $d$  steps away.
- And B is the **nearest** codeword to A, so if we cannot reach B we cannot reach **any** codeword.
- So wherever the pattern lands, it **cannot be a valid codeword**.
- The receiver sees a non-codeword and reports an error — the error is **detected**, even though it cannot be corrected.



## The rule

A code with minimum Hamming distance  $d$

- **corrects** up to  $\lfloor (d-1)/2 \rfloor$  bit errors
- **detects** up to  $d - 1$  bit errors

For the code we have been working with,  $d = 3$  — so it corrects **1** error and detects **2**.

So a good FEC code is designed to make its **minimum distance as large as possible** — the larger  $d$  is, the more errors it can correct and detect.

## RECAP

# Error correction — what we covered

**Forward error correction** repairs errors at the receiver itself, from the bits that arrived — **without retransmission**.

The **encoder** turns every  $k$ -bit data block into an  $n$ -bit **codeword**, with  $n > k$ ; the extra  $n-k$  bits are the redundancy.

The **decoder** checks whether the received pattern is a valid codeword; if it is not, an error is **detected**, and **minimum-distance decoding** returns the data block of the nearest codeword.

How many errors it can detect and correct hinges entirely on the code's **minimum Hamming distance**  $d$ .

# References

**James F. Kurose & Keith W. Ross**, *Computer Networking: a top-down approach featuring the Internet*, Addison Wesley, 3rd edition, 2005.

- Chapter 5 — The Link Layer and Local Area Network (§5.2)

**William Stallings**, *Data and Computer Communications*, Pearson, 10th edition, 2013.

- Chapter 6 — Error Detection and Correction

**Andrew S. Tanenbaum & David J. Wetherall**, *Computer Networks*, Prentice Hall, 5th edition, 2010.

- Chapter 3 — The Data Link Layer (§3.2)

# Acknowledgements

These lecture notes are developed based on slides from the following sources:

- **Dr Barry Wu**'s slides for COSC264, University of Canterbury, 2024–25
- **Dr Mengmeng Ge**'s lecture notes for COSC264, University of Canterbury, 2023
- **Assoc Prof Dan Kim**'s lecture notes for COSC264, University of Canterbury, 2017
- **Prof Aleksandar Kuzmanovic**'s lecture notes for CS340, Northwestern University, 2005

# Lineage & colophon

These lecture notes modernise COSC264 material developed by **Dr Barry Wu** (2024–25), building on notes by **Dr Mengmeng Ge** (2023) and **Assoc Prof Dan Kim** (2017), University of Canterbury, and lecture notes by **Prof Aleksandar Kuzmanovic** (CS340, Northwestern University, 2005).